

How do computers start up

1. BIOS is copied from ROM to RAM
2. BIOS start execute code
 - Initialiazes hardware
 - Runs some test POST(power-on self test)
3. Bios searches for OS to start
4. BIOS loads and starts the OS

[Operating system runs]

Legacy Booting

- BIOS loads first sector of each bootable device into memory (0x7C00)
- Bios checks for signature if found starts executing (0xAA55h)

EFI Booting

- BIOS looks into special EFI partition
- Operating System must be compiled as an EFI program

Directive

- Gives a clue to the assembler that will affect how the program gets compiled. NOT translated to machine code!
- Assembler specific
- Different assembler might have different directives.

Instructions

- Translates to machine code instructions that the cpu will execute.

Bits (directive)

-tells the assembler to emit 16/32/64-bit code.

HLT (it can be resumed by an interruption)

-stops the CPU from executing.

TIMES

-Number instructions/data

-Repeats given instruction or piece of data a number of times

\$

special symbol which is equal to the memory offset of the current line

\$\$

special symbol which is equal to the memory offset of the current sections

\$-\$\$

gives the size of our program so far

Libraries

-code oss/vs code/micro

-make

-nasm/assembler

-qemu.s